

Deep Reinforcement Learning for Trading

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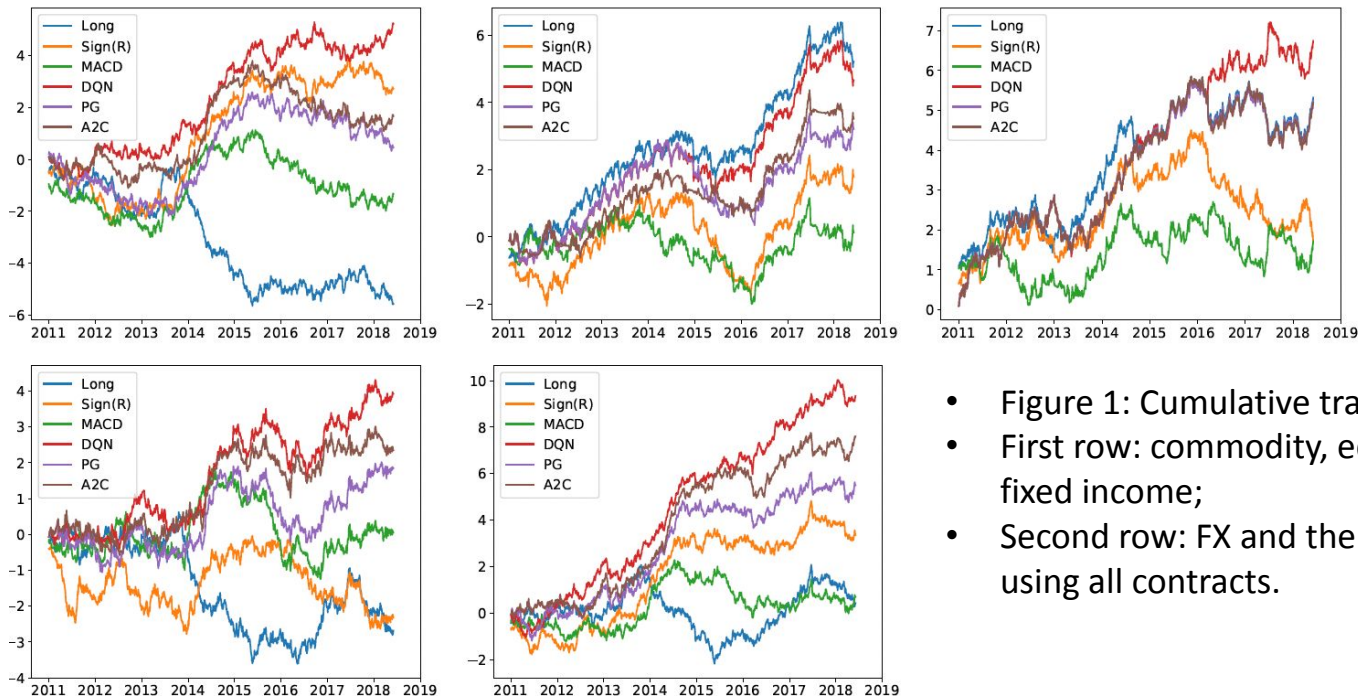
Summary of Original Paper

Traditional Methods

- Fundamental & Technical Analysis: Often lack precise timing for entering and exiting trades, leading to potential drawdowns.
- Time-Series Momentum: Highly effective in trending markets, but severely vulnerable to sideways or mean-reverting markets (like FX).
- Classical RL (Q-Learning): Frequently relies on discrete action spaces (fully long/short), which is highly risky during volatile periods.

Proposed RL Methods

- Deep Q-Networks (DQN): Evaluates the value of discrete actions in a given state.
- Policy Gradients (PG): Optimizes the policy directly, useful for continuous action spaces.
- Advantage Actor-Critic (A2C): Updates policies in real-time by combining an 'actor' network with a 'critic' network.



- Figure 1: Cumulative trade returns.
- First row: commodity, equity index and fixed income;
- Second row: FX and the portfolio of using all contracts.

Summary of Original Paper

1. Data

50 most liquid futures contracts (2011-2019).

- Commodities (25)
- Equity Indices (11)
- Fixed Income (5)
- Forex (9)

2. Methodology

Markov Decision Process (MDP) formulation.

- State: 60-day lookback (Price, Returns, MACD, RSI).
- Action: Target orders [-1, 1].
- Reward: Volatility-scaled additive profits.

3. Evaluation Metrics

Portfolio-level performance assessment.

- Annualised Expected Return
- Sharpe & Sortino Ratios
- Maximum Drawdown (MDD)
- Calmar Ratio



$$R_t = \mu \left[A_{t-1} \frac{\sigma_{tgt}}{\sigma_{t-1}} \cdot r_t - \text{bp } p_{t-1} \left| \frac{\sigma_{tgt}}{\sigma_{t-1}} A_{t-1} - \frac{\sigma_{tgt}}{\sigma_{t-2}} A_{t-2} \right| \right]$$

Table 1: Values of hyperparameters for different RL algorithms.

Model	α_{critic}	α_{actor}	Optimiser	Batch size	γ	bp	Memory size	τ
DQN	0.0001	-	Adam	64	0.3	0.0020	5000	1000
PG	-	0.0001	Adam	-	0.3	0.0020	-	-
A2C	0.001	0.0001	Adam	128	0.3	0.0020	-	-

Summary of Original Paper – Table2 & Table3

Table 2: Experiment results for the portfolio-level volatility targeting.

	E(R)	Std(R)	DD	Sharpe	Sortino	MDD	Calmar	% of + Ret	Ave. P Ave. L
Commdity									
Long	-0.710	0.979	0.604	-0.726	-1.177	0.350	-0.140	0.473	0.989
Sign(R)	0.347	0.980	0.572	0.354	0.606	0.116	0.119	0.494	1.084
MACD	-0.171	0.978	0.584	-0.175	-0.293	0.190	-0.060	0.486	1.026
DQN	0.703	0.973	0.552	0.723	1.275	0.066	0.501	0.498	1.135
PG	0.062	0.982	0.585	0.063	0.106	0.039	0.023	0.495	1.029
A2C	0.223	0.955	0.559	0.234	0.399	0.141	0.091	0.487	1.093
Equity Index									
Long	0.668	0.970	0.606	0.688	1.102	0.132	0.509	0.542	0.948
Sign(R)	0.228	0.966	0.610	0.236	0.374	0.344	0.077	0.528	0.930
MACD	0.016	0.962	0.618	0.017	0.027	0.311	0.006	0.519	0.927
DQN	0.629	0.970	0.606	0.648	1.038	0.161	0.381	0.541	0.944
PG	0.432	0.967	0.605	0.447	0.714	0.242	0.185	0.529	0.960
A2C	0.473	0.929	0.593	0.510	0.798	0.124	0.328	0.533	0.962
Fixed Income									
Long	0.680	0.975	0.576	0.698	1.180	0.061	0.444	0.515	1.054
Sign(R)	0.214	0.972	0.592	0.221	0.363	0.080	0.083	0.504	1.019
MACD	0.219	0.967	0.579	0.228	0.380	0.065	0.123	0.486	1.101
DQN	0.908	0.972	0.562	0.935	1.617	0.062	0.543	0.515	1.098
PG	0.705	0.974	0.576	0.724	1.225	0.061	0.436	0.517	1.052
A2C	0.699	0.979	0.582	0.714	1.203	0.067	0.408	0.517	1.048
FX									
Long	-0.344	0.973	0.583	-0.353	-0.590	0.423	-0.097	0.491	0.979
Sign(R)	-0.297	0.973	0.592	-0.306	-0.502	0.434	-0.111	0.499	0.954
MACD	0.006	0.970	0.582	0.007	0.011	0.329	0.002	0.493	1.029
DQN	0.528	0.967	0.553	0.546	0.955	0.183	0.313	0.510	1.051
PG	0.248	0.967	0.566	0.257	0.438	0.240	0.124	0.506	1.021
A2C	0.316	0.963	0.563	0.328	0.561	0.165	0.201	0.507	1.026
All									
Long	0.055	0.975	0.598	0.058	0.092	0.071	0.013	0.520	0.933
Sign(R)	0.429	0.972	0.582	0.441	0.737	0.038	0.201	0.510	1.031
MACD	0.089	0.978	0.582	0.091	0.153	0.008	0.035	0.493	1.043
DQN	1.258	0.976	0.567	1.288	2.220	0.002	1.025	0.543	1.043
PG	0.740	0.980	0.593	0.754	1.247	0.012	0.480	0.533	0.990
A2C	1.024	0.975	0.573	1.050	1.785	0.007	0.685	0.538	1.021

Table 3: Experiment Results for the Raw Signal.

	E(R)	Std(R)	DD	Sharpe	Sortino	MDD	Calmar	% of + Ret	Ave. P Ave. L
Commdity									
Long	-0.298	0.412	0.258	-0.723	-1.152	0.248	-0.130	0.473	0.987
Sign(R)	0.101	0.312	0.185	0.325	0.548	0.082	0.115	0.494	1.081
MACD	-0.039	0.227	0.136	-0.174	-0.290	0.132	-0.059	0.486	1.024
DQN	0.187	0.301	0.173	0.623	1.085	0.039	0.413	0.498	1.119
PG	0.013	0.287	0.172	0.047	0.078	0.072	0.017	0.495	1.026
A2C	0.072	0.163	0.098	0.440	0.729	0.099	0.161	0.487	1.151
Equity Indexes									
Long	0.504	0.928	0.606	0.543	0.831	0.127	0.466	0.541	0.928
Sign(R)	0.168	0.799	0.526	0.211	0.319	0.299	0.075	0.528	0.928
MACD	-0.068	0.586	0.385	-0.117	-0.178	0.351	-0.041	0.519	0.904
DQN	0.461	0.933	0.611	0.494	0.754	0.170	0.308	0.541	0.922
PG	0.320	0.875	0.574	0.366	0.558	0.211	0.183	0.529	0.949
A2C	0.293	0.629	0.427	0.466	0.686	0.193	0.214	0.533	0.965
Fixed Income									
Long	0.605	0.939	0.561	0.645	1.081	0.108	0.455	0.515	1.048
Sign(R)	0.189	0.795	0.496	0.237	0.381	0.165	0.103	0.504	1.024
MACD	0.136	0.609	0.367	0.224	0.371	0.124	0.131	0.485	1.102
DQN	0.734	0.862	0.508	0.851	1.445	0.118	0.469	0.515	1.086
PG	0.624	0.938	0.561	0.665	1.113	0.109	0.443	0.517	1.043
A2C	0.852	1.345	0.806	0.633	1.057	0.128	0.397	0.517	1.039
FX									
Long	-0.198	0.472	0.285	-0.420	-0.696	0.219	-0.101	0.491	0.966
Sign(R)	-0.113	0.551	0.341	-0.207	-0.332	0.170	-0.071	0.499	0.968
MACD	0.016	0.424	0.259	0.037	0.061	0.156	0.016	0.493	1.034
DQN	0.272	0.487	0.280	0.560	0.972	0.085	0.326	0.510	1.058
PG	0.157	0.533	0.312	0.295	0.503	0.098	0.148	0.506	1.029
A2C	0.159	0.455	0.267	0.349	0.592	0.081	0.193	0.507	1.034
All									
Long	-0.013	0.363	0.230	-0.036	-0.057	0.037	-0.009	0.519	0.919
Sign(R)	0.086	0.296	0.186	0.291	0.461	0.016	0.142	0.510	1.008
MACD	-0.018	0.230	0.143	-0.080	-0.129	0.026	-0.029	0.493	1.013
DQN	0.318	0.252	0.150	1.258	2.111	0.008	1.023	0.543	1.041
PG	0.168	0.279	0.174	0.602	0.968	0.011	0.373	0.533	0.968
A2C	0.214	0.221	0.134	0.969	1.601	0.009	0.672	0.538	1.014

Summary of Original Paper – Table 2 & Table 3

- With portfolio-level volatility targeting (Table 2), DRL methods (DQN, A2C) achieve the best overall risk-adjusted performance across most asset classes.
- DQN delivers the highest Sharpe and Sortino ratios in 4 out of 5 asset classes and overall, with Sharpe = 1.288 in the "All" portfolio.
- A2C performs best among baselines in Equity Index and ranks second overall.
- Without volatility scaling (raw signals, Table 3), returns are generally lower and risk-adjusted performance deteriorates, highlighting the importance of volatility targeting.
- DRL models are able to capture large trends while scaling down or holding during consolidation periods, generating positive profits after transaction costs.

Reproduction Results

Table 2: Experiment results for the portfolio-level volatility targeting

Additional portfolio-level vol scaling applied (target $\sigma = 0.97$ annualised).

Model	E(R)	Std(R)	DD	Sharpe	Sortino	MDD	Calmar	% +Ret	Ave.P/Ave.L
Commodity									
Long	-0.760	0.970	0.662	-0.784	-1.149	8.744	-0.087	0.479	0.958
Sign(R)	-0.406	0.970	0.645	-0.418	-0.629	4.986	-0.081	0.494	0.956
MACD	-0.829	0.970	0.646	-0.854	-1.283	8.294	-0.100	0.487	0.913
DQN	-0.935	0.970	0.627	-0.964	-1.492	9.269	-0.101	0.480	0.923
A2C	-0.628	0.970	0.656	-0.647	-0.957	6.802	-0.092	0.497	0.910
Equity Index									
Long	0.547	0.970	0.752	0.564	0.728	1.924	0.284	0.546	0.915
Sign(R)	-0.034	0.970	0.756	-0.035	-0.044	3.355	-0.010	0.520	0.918
MACD	-0.515	0.970	0.744	-0.531	-0.692	6.024	-0.085	0.500	0.908
DQN	-0.345	0.970	0.697	-0.356	-0.496	5.482	-0.063	0.460	1.096
A2C	0.503	0.970	0.797	0.518	0.630	1.720	0.292	0.548	0.904
Fixed Income									
Long	0.628	0.970	0.644	0.647	0.974	1.809	0.347	0.534	0.969
Sign(R)	0.025	0.970	0.674	0.026	0.037	3.801	0.007	0.512	0.958
MACD	-0.338	0.970	0.652	-0.349	-0.519	6.286	-0.054	0.467	1.076
DQN	0.004	0.970	0.904	0.004	0.005	3.679	0.001	0.261	1.034
A2C	0.493	0.970	0.633	0.508	0.779	2.206	0.224	0.523	0.993
FX									
Long	-0.397	0.970	0.625	-0.409	-0.634	5.454	-0.073	0.490	0.972
Sign(R)	-0.778	0.970	0.682	-0.802	-1.141	8.248	-0.094	0.478	0.952
MACD	-0.927	0.970	0.688	-0.955	-1.347	9.256	-0.100	0.465	0.971
DQN	-1.522	0.970	0.795	-1.569	-1.913	14.366	-0.106	0.368	0.919
A2C	-0.547	0.970	0.631	-0.563	-0.867	6.314	-0.087	0.487	0.959
All									
Long	-0.037	0.970	0.675	-0.038	-0.055	3.914	-0.010	0.515	0.934
Sign(R)	-0.431	0.970	0.692	-0.444	-0.623	5.416	-0.080	0.494	0.951
MACD	-1.073	0.970	0.686	-1.106	-1.564	10.908	-0.098	0.467	0.943
DQN	-1.256	0.970	0.672	-1.294	-1.868	11.875	-0.106	0.457	0.953
A2C	-0.090	0.970	0.679	-0.093	-0.133	3.882	-0.023	0.515	0.928

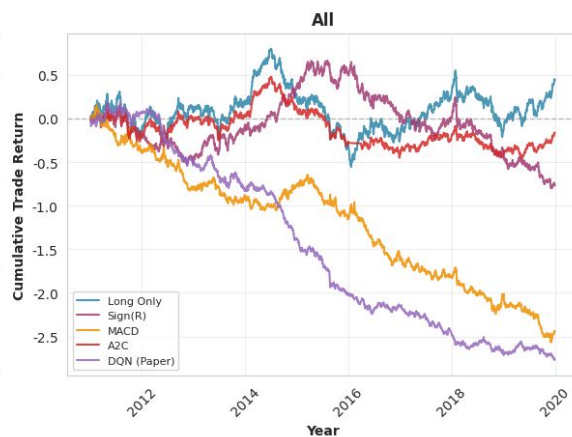
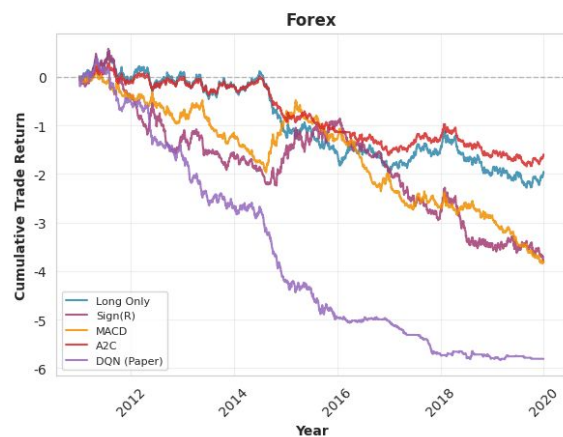
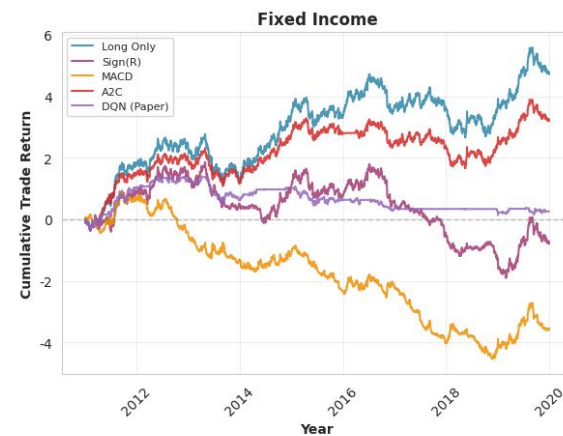
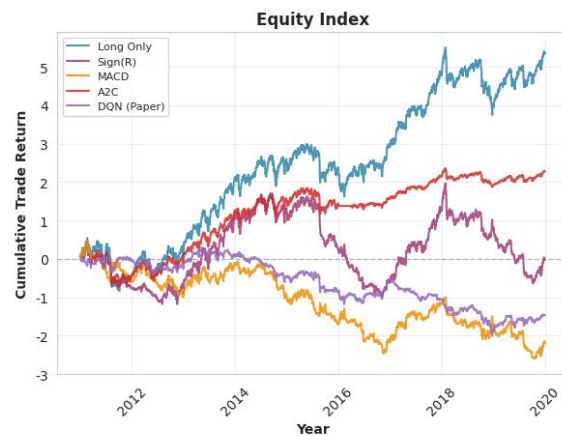
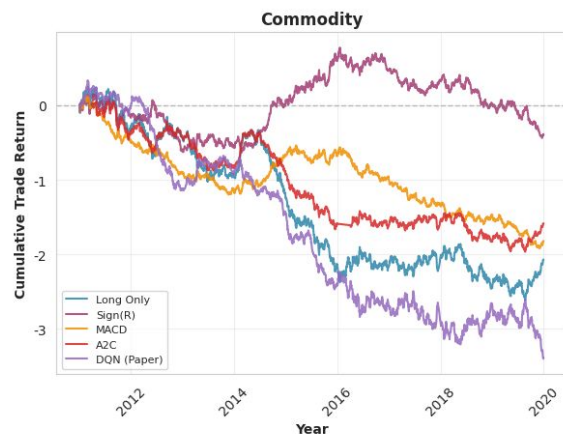
Table 3: Experiment Results for the Raw Signal

No additional portfolio-level vol scaling (raw per-contract signal only).

Model	E(R)	Std(R)	DD	Sharpe	Sortino	MDD	Calmar	% +Ret	Ave.P/Ave.L
Commodity									
Long	-0.323	0.413	0.281	-0.784	-1.149	3.719	-0.087	0.479	0.958
Sign(R)	-0.136	0.324	0.216	-0.418	-0.629	1.667	-0.081	0.494	0.956
MACD	-0.212	0.248	0.165	-0.854	-1.283	2.118	-0.100	0.487	0.913
DQN	-0.240	0.249	0.161	-0.964	-1.492	2.383	-0.101	0.480	0.923
A2C	-0.186	0.288	0.194	-0.647	-0.957	2.016	-0.092	0.497	0.910
Equity Index									
Long	0.510	0.905	0.701	0.564	0.728	1.794	0.284	0.546	0.915
Sign(R)	-0.026	0.760	0.592	-0.035	-0.044	2.627	-0.010	0.520	0.918
MACD	-0.322	0.606	0.465	-0.531	-0.692	3.762	-0.085	0.500	0.908
DQN	-0.199	0.558	0.401	-0.356	-0.496	3.155	-0.063	0.460	1.096
A2C	0.257	0.496	0.408	0.518	0.630	0.880	0.292	0.548	0.904
Fixed Income									
Long	0.603	0.933	0.619	0.647	0.974	1.739	0.347	0.534	0.969
Sign(R)	0.021	0.801	0.556	0.026	0.037	3.138	0.007	0.512	0.958
MACD	-0.228	0.653	0.438	-0.349	-0.519	4.228	-0.054	0.467	1.076
DQN	0.001	0.239	0.223	0.004	0.005	0.906	0.001	0.261	1.034
A2C	0.364	0.717	0.467	0.508	0.779	1.629	0.224	0.523	0.993
FX									
Long	-0.188	0.460	0.296	-0.409	-0.634	2.584	-0.073	0.490	0.972
Sign(R)	-0.436	0.544	0.382	-0.802	-1.141	4.622	-0.094	0.478	0.952
MACD	-0.421	0.440	0.312	-0.955	-1.347	4.201	-0.100	0.465	0.971
DQN	-0.459	0.293	0.240	-1.569	-1.913	4.331	-0.106	0.368	0.919
A2C	-0.184	0.327	0.213	-0.563	-0.867	2.128	-0.087	0.487	0.959
All									
Long	-0.014	0.356	0.248	-0.038	-0.055	1.435	-0.010	0.515	0.934
Sign(R)	-0.132	0.298	0.212	-0.444	-0.623	1.663	-0.080	0.494	0.951
MACD	-0.275	0.248	0.176	-1.106	-1.564	2.792	-0.098	0.467	0.943
DQN	-0.234	0.181	0.125	-1.294	-1.868	2.216	-0.106	0.457	0.953
A2C	-0.022	0.235	0.165	-0.093	-0.133	0.942	-0.023	0.515	0.928

Reproduction Results

Strategy Comparison: Long Only / Sign(R) / MACD / A2C / DQN (Paper)
(2011-01-01 to 2019-12-31)



Differences vs Original

- Original paper: DQN and A2C generate strong positive cumulative returns in most asset classes, especially Commodity, Fixed Income, FX, and All portfolio.
- Our reproduction: RL models do not show stable alpha. DQN often trends downward and underperforms simple baselines.
- Equity Index is the only asset class where the reproduced pattern is broadly consistent with the paper: Long Only performs best due to the strong upward equity market trend.
- Commodity and FX show the largest discrepancy. The paper claims RL can handle volatile or mean-reverting markets, but our reproduced DQN suffers large cumulative losses.
- All portfolio fails to show the diversification benefit reported in the paper. The reproduced DQN remains negative even after combining all contracts.

Reasons for Differences

1. Data quality issues

- Three contracts contained only missing values.
- Some RAD files showed abnormal price jumps.

2. Ambiguous LSTM-RL implementation

- The paper states that LSTM networks are used for the RL models, but the exact implementation is not fully specified.
- It is unclear how LSTM is integrated into DQN and A2C.

3. Missing key hyperparameters

- The paper does not fully specify some important parameters required for exact reproduction. The most important missing parameter is the **sigma target** used in portfolio-level volatility scaling.
- Volatility scaling directly affects rewards, position sizes, and cumulative returns.

4. Original Figure 1 has visual inconsistencies

- The cumulative return curves in the original Figure 1 do not appear to start from the same level across all asset classes. This suggests that additional normalization or plotting adjustments may have been applied.

Robustness Checks

1. Train/Test Period Robustness

We did two train/test split following original paper to see whether the A2C strategy remains effective across different market regimes and out-of-sample periods.

- Train: 2005–2010, Test: 2011–2015
- Train: 2005–2015, Test: 2016–2019

2. Asset Class Robustness

Performance is evaluated separately for Commodity, Equity Index, Fixed Income, FX / Forex, All Contracts. This checks whether the strategy performance is consistent across different futures markets rather than being driven by a single asset class.

3. Transaction Cost Robustness

The trained A2C signals are re-evaluated under different transaction cost assumptions without retraining the model (1bp, 5bp, 10bp, 15bp, 20bp, 25bp, 30bp, 35bp, 40bp, 45bp). This checks whether the strategy remains profitable under more conservative execution cost assumptions.

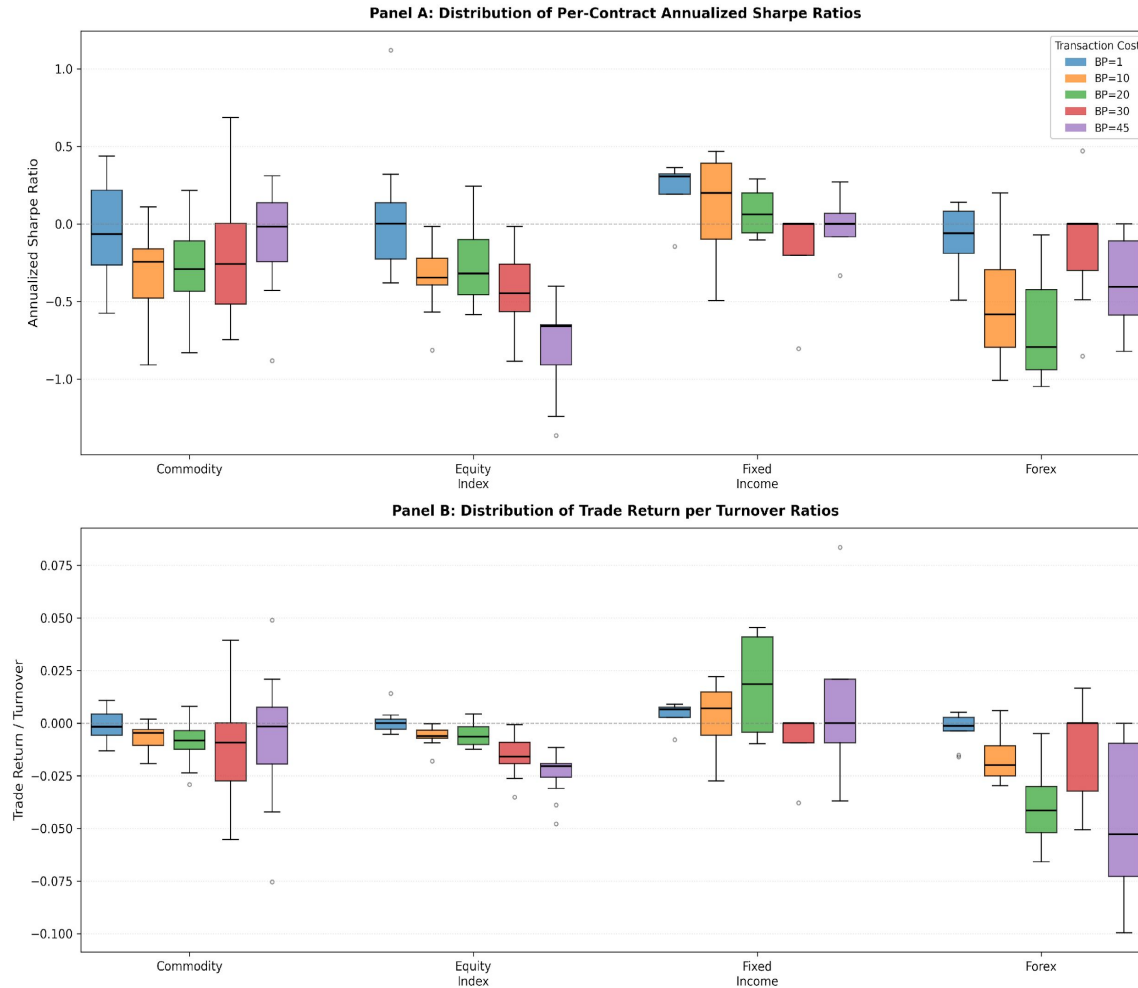
4 One deep dive:

For DQN, we did a deep dive on contracts and retrained through 5 transaction costs level(1/10/20/30/45) across 10 seeds(42-51)

- Transaction Costs Actively Shape Learning, Not Just Suppress Performance
- Different Asset Classes Respond Differently to Cost Changes
- Trading Activity Shows Non-Monotonic, Adaptive Behavior

Transaction Cost Robustness: per-contract Sharpe(DQN)

Exhibit 4: Per-Contract Performance Analysis by Transaction Cost



Fixed Income holds up best

Sharpe stays positive through 20 bps, suggesting lower turnover sensitivity and more stable signal quality.

Equity Index / Forex are cost-sensitive

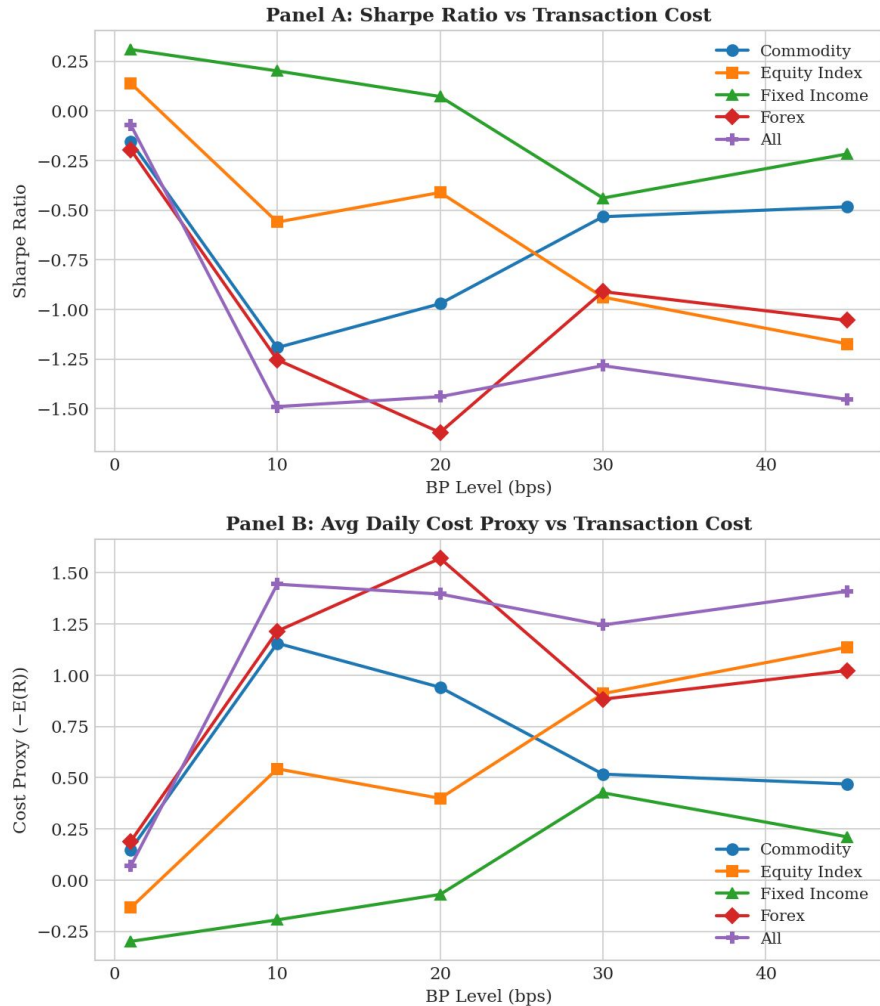
Performance deteriorates faster as bps rise, so the learned policy is less robust to realistic execution frictions.

Not a simple linear penalty

The non-monotonic BP pattern indicates strategy adaptation, not just a mechanical subtraction from returns.

Transaction Cost Robustness: fee level changes behavior

Exhibit 5: Transaction Cost Impact on DQN Performance



Learning process changes

Higher fees reduce the share of positive trades faster than average P/L, consistent with fewer, larger bets replacing many small trades.

Daily cost scales unevenly by asset

Equity Index and Forex absorb the largest realized costs at high BP levels, while Fixed Income starts from a much lower base.

Sharp break after 1 bp

All-contract Sharpe falls from near flat at 1 bp to about -1.49 at 10 bps, showing cost assumptions matter a lot whether agents really gain the edge through learning

Key Takeaways

1. Transaction Costs Actively Shape Learning, Not Just Suppress Performance

The seed ranking inversions (Seed 45 going from rank 8.33 at BP=1 to rank 3.0 at BP=20) prove that transaction costs determine which strategy/seed succeeds, not just how well a single strategy performs. This is a fundamental insight for RL in financial markets.

- Seed 45: BP=1 rank 8.33 → BP=20 rank 3.0
- Seed 42: BP=1 rank 4.25 → BP=20 rank 7.5 → BP=45 rank 3.5
- Strategy rankings are not preserved across cost levels

2. Different Asset Classes Respond Differently to Cost Changes

Fixed Income maintains profitability across cost levels (positive Sharpe through BP=20), while Equity Index and Forex show extreme sensitivity.

- Fixed Income Sharpe: +0.309 (BP=1) → +0.072 (BP=20) → -0.217 (BP=45)
- Equity Index Sharpe: +0.139 (BP=1) → -0.411 (BP=20) → -1.173 (BP=45)
- Forex Sharpe: -0.195 (BP=1) → -1.619 (BP=20) → -1.055 (BP=45)

3. Trading Activity Shows Non-Monotonic, Adaptive Behavior

Agents don't simply trade less as costs increase. They show complex patterns including increased activity at BP=10, near-withdrawal at BP=30, and partial recovery at BP=45.

- Commodity: 5512 → 5954 → 4161 → 277 → 686 trading days
- Total trading days: 11652 → 11054 → 8001 → 1098 → 2509
- 91% reduction in trading at BP=30, then 128% increase at BP=45

Paper View	Our Finding
Costs reduce all strategies proportionally	Costs favor different seeds at different levels
Ranking of seed is preserved	Rankings invert (Seed 45: worst → best)
Trading activity decreases monotonically	Trading shows U-shaped recovery (BP=30 → BP=45)
One seed is universally optimal	Optimal seed depends on cost environment

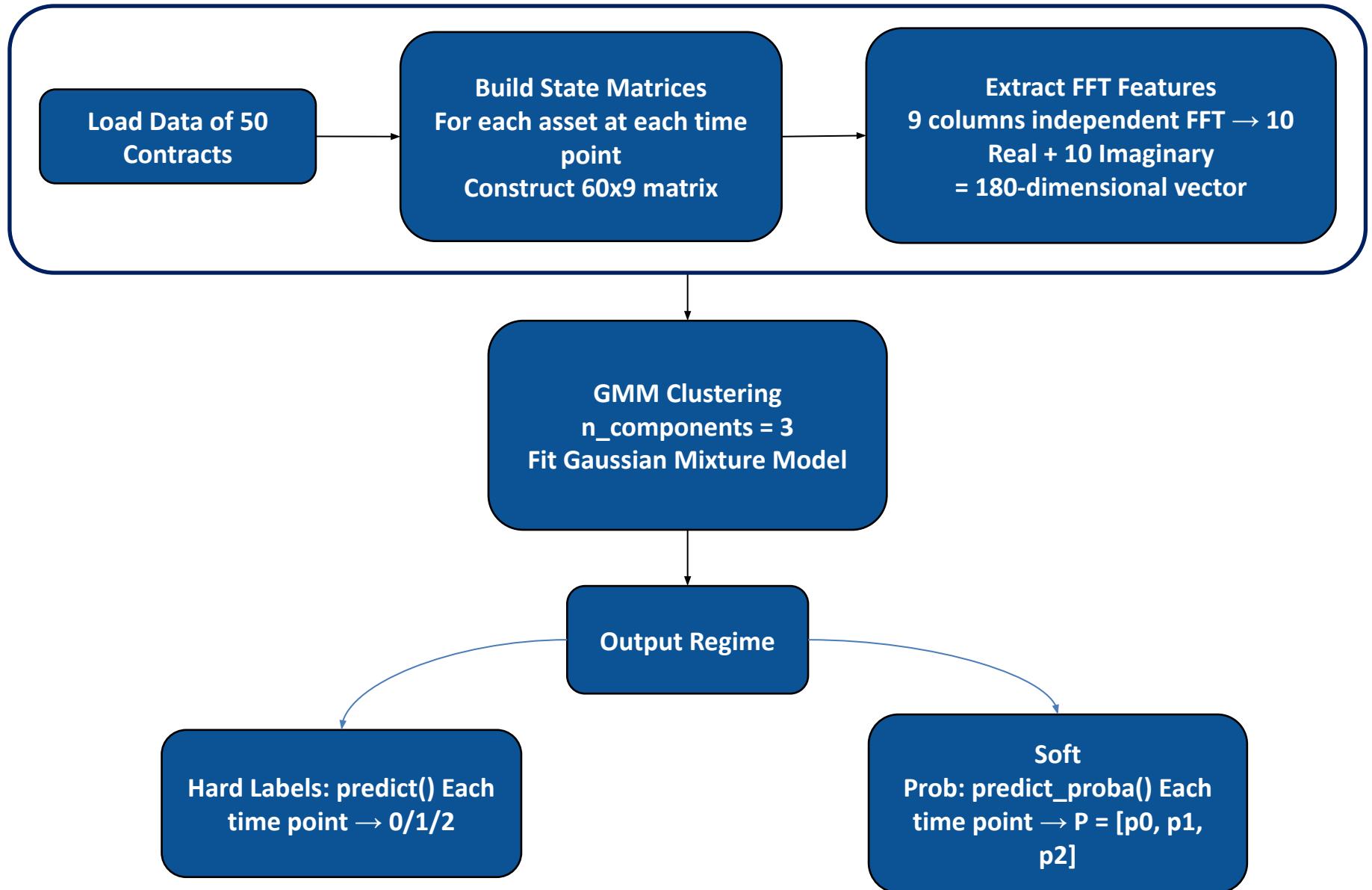
Risk diagnostics

	Long	Sign(R)	MACD	DQN	A2C
Commodity					
E(R)	-0.760	-0.406	-0.829	-0.935	-0.628
Sharpe	-0.784	-0.418	-0.854	-0.964	-0.647
Equity Index					
E(R)	0.547	-0.034	-0.515	-0.345	0.503
Sharpe	0.564	-0.035	-0.531	-0.356	0.518
Fixed Income					
E(R)	0.628	0.025	-0.338	0.004	0.493
Sharpe	0.647	0.026	-0.349	0.004	0.508
FX					
E(R)	-0.397	-0.778	-0.927	-1.522	-0.547
Sharpe	-0.409	-0.802	-0.955	-1.569	-0.563
All					
E(R)	-0.037	-0.431	-1.073	-1.256	-0.090
Sharpe	-0.038	-0.444	-1.106	-1.294	-0.093

Source: Reproduced results (Table 2, portfolio-level volatility targeting).

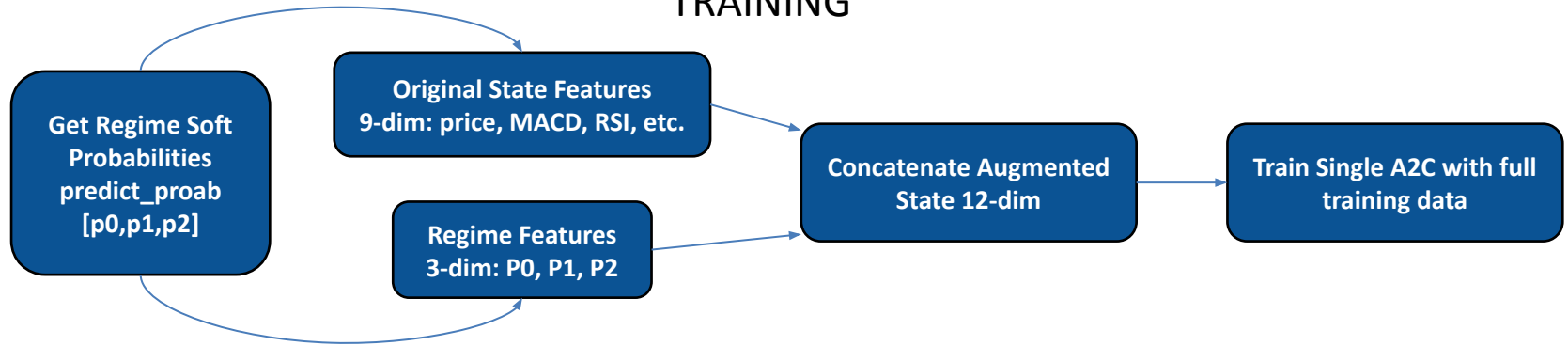
The diagnostic table shows that Sharpe performance is mainly driven by E(R), since volatility is fixed around the portfolio target. Positive Sharpe only appears when the strategy generates positive expected returns, mainly in Equity Index and Fixed Income. In contrast, Commodity, FX, and the All portfolio show mostly negative E(R) and Sharpe, suggesting weak return generation rather than excessive volatility as the main source of underperformance.

Extension - Regime Detection:

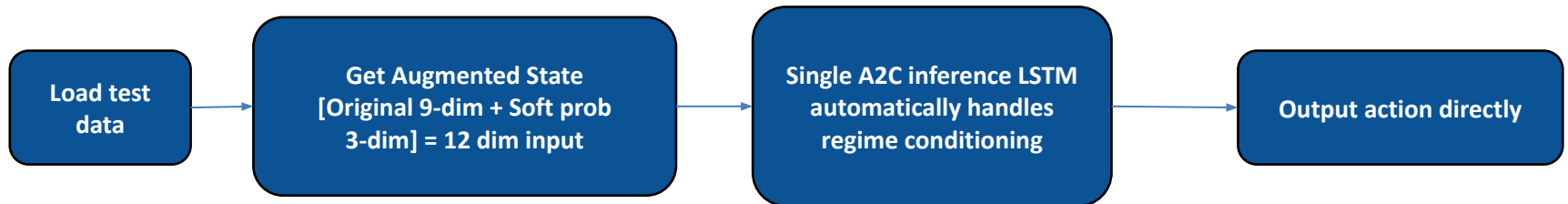


Extension - Regime Detection:

TRAINING

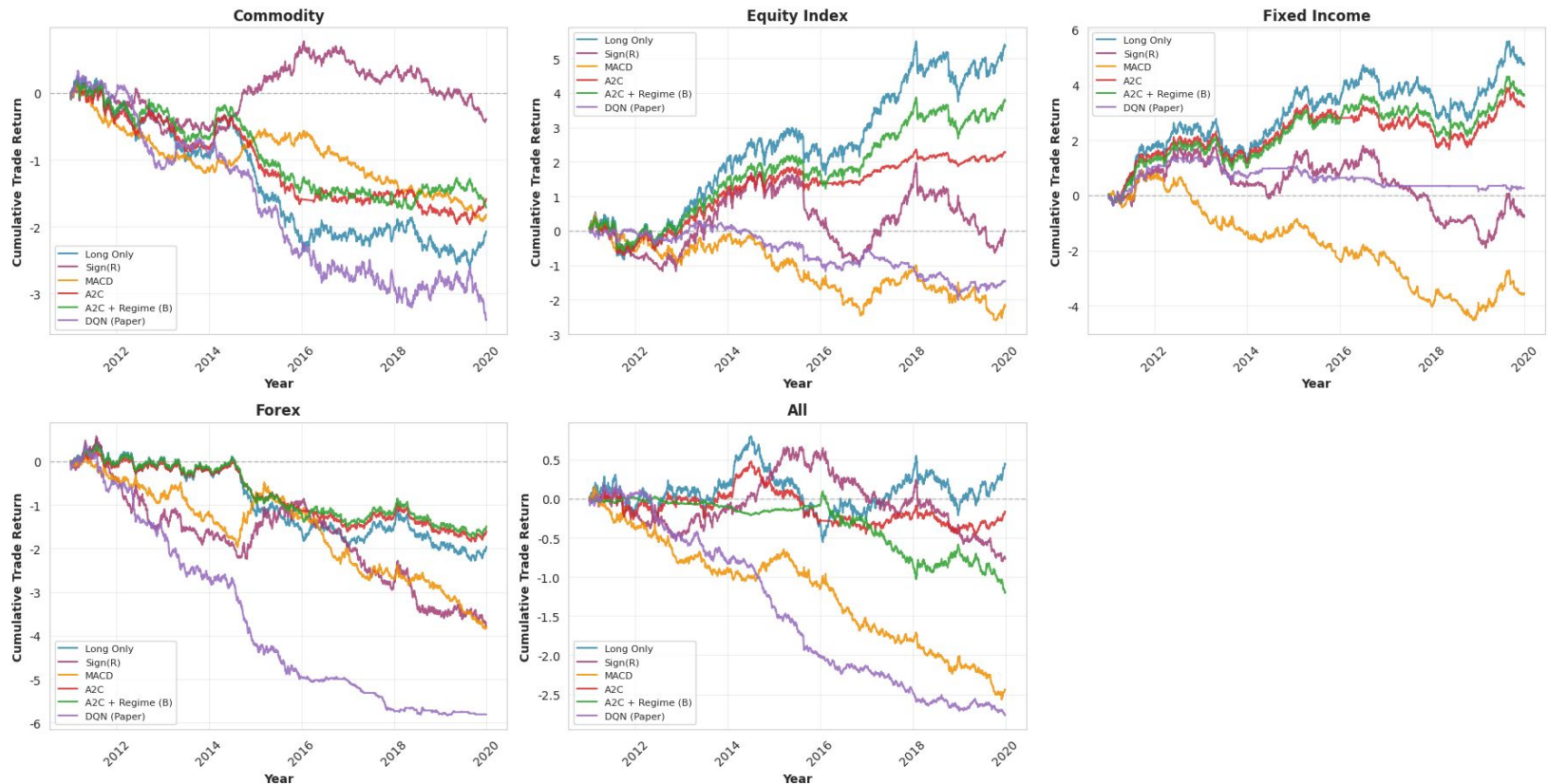


TEST



Extension - Regime Detection:

Strategy Comparison: Long Only / Sign(R) / MACD / A2C / A2C+Regime (Route B) / DQN (Paper)
(2011-01-01 to 2019-12-31)



A2C + Regime Detection improves performance in Equity Index and Fixed Income, where market trends are more persistent and easier to identify. **The improvement is limited in Commodity and FX**, where the adjusted A2C still produces negative cumulative returns. **At the All portfolio level, the benefit is not robust**: the regime-adjusted A2C becomes unstable after 2016 and eventually trends downward.

References

1. Zhang, Z., Zohren, S., & Roberts, S. (2020). Deep Reinforcement Learning for Trading. *Journal of Financial Data Science*, 2(2), 25-40. <https://doi.org/10.3905/jfds.2020.1.030>
2. Moskowitz, T. J., Ooi, Y. H., & Pedersen, L. H. (2012). Time Series Momentum. *Journal of Financial Economics*, 104(2), 228-250. <https://doi.org/10.1016/j.jfineco.2011.11.003>
3. Baz, J., Granger, N., Harvey, C. R., Le Roux, N., & Rattray, S. (2015). Dissecting Investment Strategies in the Cross Section and Time Series. SSRN. <https://doi.org/10.2139/ssrn.2695101>